IIT Kharagpur

The RuPayPay Crisis : Scaling at a Crossroad

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A Comprehensive Overview of RupayPay and the Problem

Understanding the Business Models - RuPayPay vs Paytm & BharatPe

RuPayPay Credit-Led Model

Merchant Onboarding
Acquire underserved merchants via payment gateway from Tier2/3 SMEs.

Data-Driven Credit Scoring
Uses GST + transaction data for instant, low-risk underwriting.

Working Capital Loans as Core Revenue
Loan book ₹320 Cr, NPAs only 2.1%

Loans=Profits
Payments lock in users; feedback loop improves credit model.

Paytm & BharatPe Payment-Led Model

Mass Merchant Acquisition
5k–10k merchants/day via QR codes, discounts, incentives.

Payments at Scale (Low/Zero MDR) → GMV growth prioritized; razor-thin margins.

Lending as Cross-Sell
Loans offered to lock in merchants; NPAs rise (3.5–5%).

Growth over Profitability
Bundle wallets, BNPL, insurance;

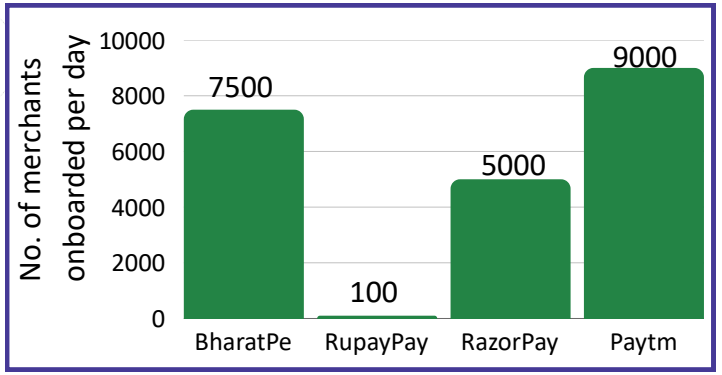
SWOT Analysis of RupayPay

Strengths

- **Proprietary GST** + transaction-data credit model keeps **NPAs at 2.1% (vs 3.5–5% industry).**
- Bootstrapped yet profitable (**12% EBITDA margin**) with **80,000 loyal SME customers**. High **CAGR** OF **70%** with **52 Cr** in Revenue

Weaknesses

- Growth limited by **capital constraints** (only ~100 merchants/day vs 5k–10k for peers).
- Smaller **brand visibility** vs Paytm, BharatPe, Razorpay.



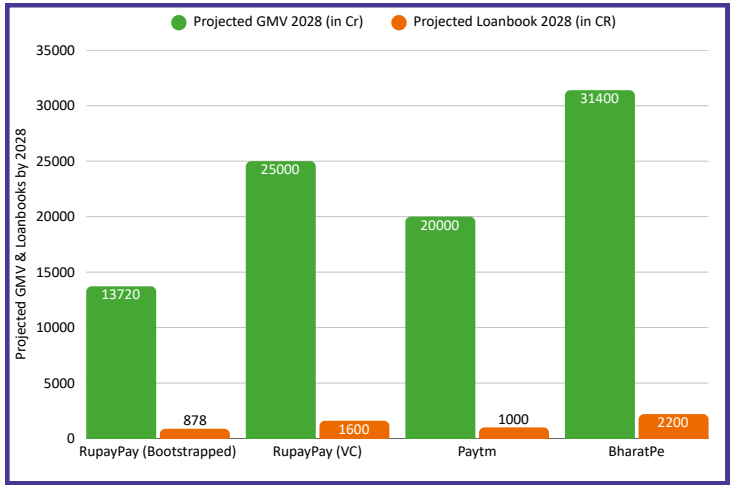
Thus, the **key question** before the us is: how can RupayPay balance the **urgency to scale with the risks of growth,** and should that journey be fueled by **venture capital** like other category leaders or by its proven **bootstrapped discipline**?

Opportunities

- **$240–360B SME credit gap** with <16% served by banks = huge untapped market.
- UPI penetration + **Tier II/III expansion** enable lending + **cross-sell opportunities.**

Threats

- **Regulatory tightening** (RBI FLDG, zero-MDR, compliance burden).
- Rivals with **larger war chests** may capture "category captain" positions, if preparation for the future is not begun immediately.



Scenario Models

		GMV (₹ cr)	Loanbook (₹ cr)	Customer Count	Net yield (%)	NII (₹ cr)	Payment Rev (₹ cr)	Total Revenue (₹ cr)	Opex (₹ cr)	Credit Cost (₹ cr)
VC Fast Scale	Year 1	8000	512	1280000	11%	56.32	26.88	83.2	70	30.72
	Year 3	25000	1600	4000000	11%	176	84	260	188	72
Boostrapping	Year 1	7000	448	1120000	11%	49.28	23.52	72.8	52.4	9.408
	Year 3	13720	878.08	2195200	11%	96.5888	46.0992	142.688	98.3	15.80544

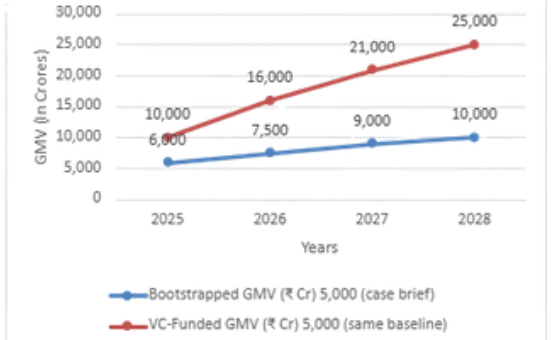
		EBITDA (₹ cr)	NPA %	NPA Amount (₹ cr)	Capex (₹ cr)	Cash Burn (₹ cr)
VC Fast Scale	Year 1	-17.52	6%	30.72	35	52.52
	Year 3	0	4.50%	72	0	0
Boostrapping	Year 1	10.992	2.10%	9.408	5	5
	Year 3	28.58256	1.80%	15.80544	0	0

Assumptions

- **Baseline GMV** (₹ cr) = 5,000
- **Loanbook/GMV ratio** = 6.4%
- Average outstanding per customer = ₹40,000
- **Gross loan yield** = 22%
- **Cost of funds** = 11%
- **Payment take rate** = 33.6 bps (0.00336)

- **Provision rate / Credit cost %** = scenario-specific (e.g., 6% for VC Y1, 2–3% for Bootstrap Y1)
- **CapEx** (Year 1) = scenario-specific (₹35 cr for VC, ₹5 cr for Bootstrap)
- **Opex** (Year 1) = given (₹52 cr Bootstrap, ₹70 cr VC)

Projected GMV of RupayPay across 3 years



Why consider VC?

With 0% MDR, payments do not generate revenue and lending must drive monetization.

- In the bootstrap path, the loanbook reaches only ₹878 cr by Year 3, revenues remain too small and payments become a liability.
- In the VC path, the loanbook grows to ₹1,600 cr by Year 3, creating strong NII and cross-sell so payments turn into sticky funnels.

Only at VC scale do "free" UPI payments become profitable.

2X

- VC path doubles GMV and loanbook scale vs bootstrap.
- Nearly 2× revenue potential (₹260 cr vs ₹143 cr).
- Larger merchant base (4.0 lakh vs 2.2 lakh) providing stronger network effects.
- Higher upfront burn (₹53 cr vs ₹5 cr) is the price of building category leadership.
- Bootstrap shows short-term EBITDA, but VC delivers much higher long-term enterprise value.

Evaluation of Alternatives and Risk Mitigation Strategies

Rewards

Bootstrapped

- **Financial Strength:** Maintain 12% EBITDA margin → consistent profitability, self-sustained growth.
- **Control & Autonomy:** Founders retain 100% equity and decision-making power

VC Funded

- **Rocket-Fuel Growth**: GMV can scale to ~₹25,000 Cr; loan book ₹4,500–5,000 Cr; revenue >₹600 Cr by FY28, Govt. Subsidies. Source: Redseer 2024
- **Strategic Edge:** Funds enable investments in AI underwriting, compliance, regulatory readiness, and brand-building

So,**Bootstrapping = profitable irrelevance.**
VC path = survival through scale, if executed with discipline.
Therefore, we recommend **VC funding of ₹125 Cr**

Risks (3 Year Analysis)

	Bootstrapped	VC Funded
Financial Risk	GMV capped at ~₹10,000 Cr; loan book ~₹1,000–1,200 Cr; revenue ~₹150 Cr by FY28 (vs 70% CAGR historically). *Source:* IFC MSME Credit Gap data	Scales GMV to ₹25,000 Cr, loan book ~₹5,000 Cr, revenue >₹600 Cr by FY28; short-term EBITDA negative (–15% to –20%) Source: Redseer report
Operational Risk	100 merchants/day vs Paytm/BharatPe's 5k–10k/day Paytm GMV >₹50,000 Cr vs RupayPay ~₹10,000 Cr by 2028. (Projected)	NPAs may rise to 3.5–4% (Projected as per RBI Benchmarks) Source:Economic Times
Cultural Risk	Team stays lean but risks stagnation and talent attrition.	Hiring ≈5–7× risks cultural dilution unless mitigated via structured onboarding.
Regulatory Risk	NPAs stable at ~2.1–2.5%, however, limited resources to invest in compliance readiness.	RBI zero-MDR + FLDG pressures; allocating ₹20 Cr to compliance is necessary (RBI zero-MDR regulation)

Three Must-Win Battles for VC Approach

- **Credit Control:** Maintain **NPAs ≤2.5%** via GST-data underwriting, vintage monitoring, and collections.
- **Regulatory Readiness**: Invest in **NBFC compliance, FLDG buffers, and digital KYC infrastructure**.
- **AI/Analytics & Hiring**: Build AI underwriting/fraud systems and **scale from 74 → 500** employees using structured culture onboarding.

Risk Mitigation & Contingency Plans for a VC Funded Approach

- **Funding-linked Milestones**: Tranches tied to performance and NPA thresholds.
- **Operational Safety:** Pilot new cities with incremental rollout; delay scaling lending until repayment data is stable.
- **Cultural Stability:** Maintain lean DNA through founder-led governance teams.

- **Equity Ask:** Prefer ₹100 Cr for 15% stake in tranches
- **Control Safeguards:** Founders retain veto over credit, hiring, and regulatory decisions; and retain at least 2/3 board seats
- **Cost of Capital Control:** Ensure cost of funds ≤9% to protect spread.
- **Independent Oversight:** Quarterly audits

Recommendations

Ideas Inspired from Global Standpoint

China – MYbank (Ant Group)

- "3–1–0" lending model: 3 min to apply, 1 sec to approve, 0 manual steps → instant SME credit.
- Cloud-native platform enabling low-cost, scalable lending.
- Uses alternative data (e-commerce sales, payments, tax filings) to underwrite thin-file SMEs.

Outcome: Rapid, mass-market SME credit with controlled risk.

Cost Breakdown

- **Phase 1** (Infra & Tech, 0–6m): ₹35–50 cr for AI underwriting, compliance, and data stack.
- **Phase 2** (3-4 Cities, 6–18m): ₹35-45 cr for marketing, local teams, and loan provisioning.
- **Phase 3** (Scale to 10 Cities, 18–36m): ₹45-55 cr for operations and product expansion.

Total 3-Year Spend: ₹115–150 cr, well-covered by the ₹125 cr raise.

**The Whole Cost breakdown given in appendix*

NEGOTIATION- For Equity from 20% to 15%

Industry Growth Premium

- 70% CAGR vs industry avg 25–30% → deserves higher multiple.
- 6–8× revenue = ₹364–416 cr valuation.
- Forward (3 yrs): ₹2,600–3,900 cr → 20% for ₹125 cr undervalues upside.

Strong Asset Quality

- Lower risk gives higher valuation multiple on loan book.
- NPA = 2.1% vs industry avg 2.5–3%.
- Lower risk = premium loan book valuation (1.2× ₹320 cr = ~₹384 cr).

Customer Base & Efficiency

- 80k SMEs → sticky ecosystem.
- 74 employees with ₹70L revenue/employee vs fintech avg ₹35–40L.
- High efficiency = strong scalability.

Valuation Math @ 15%

- If the investor puts ₹125 cr for 15%, the post-money valuation is ₹833 cr and pre-money is ₹708 cr.
- This is a 30–40% premium over the VC's ₹500 cr pre-money.
  The premium is justified by strong growth, a large SME base, and low NPAs.

Scenario Check

- 3 yrs @ GMV ₹25,000 cr → Revenue ≈ ₹260 cr, EBITDA ≈ ₹39 cr.
- Valuation @10× revenue ≈ ₹2,600 cr.
  VC at 20% → stake ≈ ₹520 cr (4× return).
- VC at 15% → stake ≈ ₹390 cr (3× return).
- Both deliver strong 3–4× returns → fair to negotiate at 15%.

Category	VC Offer: ₹125 cr for 20% (Base Case)	Scenario 2: Negotiated Terms (Proposed)
Financial Terms	• Pre-Money Valuation: **₹500 cr** • Post-Money Valuation: **₹625 cr** • Equity Dilution: **20%**	• Pre-Money Valuation: **₹708 cr** • Post-Money Valuation: **₹833 cr** • Equity Dilution: **15%**
VC "Ask"	• GMV: **₹25,000 cr in 3 yrs** • Enter **10 new cities** • Series B readiness in **18–24 months**	• GMV: **₹15,000–18,000 cr in 3 yrs** (sustainable) • Enter **3–4 cities/yr → 10 cities in 3 yrs (phased)** • Series B readiness in **24–30 months**
Strategic Fit	• **High risk, high reward** • May overstretch infra → NPAs could rise to **6%** • Execution pressure may dilute customer experience	• **Balanced risk-return** • Controlled rollout keeps NPAs ≤ **3–4%** • Better alignment with current economic contraction cycle
Mitigation	• Ask for **stage-based capital release** tied to metrics (e.g., PAR30, CAC, Loan book quality) • Reframe success metrics: NPAs < 4%, EBITDA > 10%, CAC payback < 6 months	• Back proposal with **data-driven model**: 70% CAGR revenue, low NPAs (2.1% vs 3% avg), 80k SME base • Highlight **global analogues** (M-Shwari, MYbank) that scaled through phased growth • Show VC still gets **3–4× returns in 3 yrs**

Roadmap

Phase 1- Foundation Building (Contraction Phase - Approx. 0-9 Months)

Objective: Build core tech, data stack, compliance and hiring so scale later does not blow NPAs or CAC.

Key Activities

- **Underwriting Engine:** Develop/subscribe AI/ML-driven system (₹5–15 cr; SaaS ₹2–5 cr).
- **Data Infra & Integrations:** Scalable platform with UPI, GSTN, AA linkages (₹2–5 cr).
- **Payments & Compliance:** Upgrade gateway for instant disbursal; strengthen compliance & cybersecurity (₹1–2 cr).
- **Core Team Build:** Hire risk lead, data scientists, compliance head (₹5–10 cr)

Cost (Range)

Tech / infra (one-time): ₹16–39cr (per your Table3 low/high).
Hiring + initial ops (Y1): additional ₹15–20cr
Total Phase 1 spend: ₹35–55cr (matches earlier scenario CapEx & initial opex)

Phase 2- Strategic Launch (Early Recovery Phase - Approx. 9-18 Months)

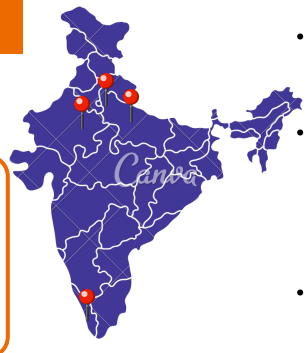
Key Activities

- **Lending in 3-4 cities:** Offer small-ticket SME loans (₹10–50k), repayments linked to UPI flows.
- **Partnerships:** Tie up with 1–2 anchor marketplaces per pilot city for adoption.
- **Live Scorecard:** Build credit models from UPI/GST & repayment signals.
- **Ops Refinement:** Strengthen fraud detection, risk monitoring, and customer onboarding.

Cost (Range)

Pilot marketing / field ops / hires / loan provisioning: ₹40–60cr over Phase2 (includes capital to fund loan book increase and marketing to acquire 20k–30k new merchants).
Total Phase2 incremental spend (9–18m): ₹35-45cr.

3 Cities-



- **Jaipur**- Consistently leads Tier-2 cities in new business incorporations and MSME activity, topped quarterly incorporations with 1,678 in Q4 FY25.
- **Lucknow**- Major trading center for Northern India; saw high MSME entrepreneurial activity and a 91% surge in loan applications among Tier 2/3 cities. Local merchants are increasingly seeking credit for business expansion and inventory finance.
- **Ghaziabad**- Delhi NCR's key wholesale/logistics hub. Led in Tier-2 business incorporations (1,053 in Q4 FY25) and shows robust credit demand post-downturns

**Details in the appendix*

Phase 3 - Accelerated Expansion (Late Recovery/ Expansion Phase - Approx. 18-24 Months)

Objective: Leverage the proven infrastructure and market presence for rapid, data-driven growth.

Key Activities

- **City Rollout:** Expand to remaining cities in the roadmap, using refined processes from Phase 2.
- **Product Suite Expansion:** Add invoice discounting, merchant overdrafts, BNPL
- **Operational Strengthening:** Regional risk desks, collections, customer success teams for scale.
- **Growth Focus:** Aggressively drive GMV & loan book growth while keeping NPAs ≤ 3–4%.
- **Funding Prep**: Position for Series B within 24–30 months with validated growth trajectory.

Cost (Range)

Rollout ops + product dev + regional hires: ₹40–70cr across Phase3 (this includes working capital for loan book growth, marketing and city-level operating budgets).

Total Phase3 spend = ₹45-55 cr

Appendix

TOPIX
-0.69
Close 1,196.78
+25.6
Close 14,467.14
WEEKLY CLOSES
HONG KONG, SHANGHAI
HONG KONG (AFP-Jiji)—Ho
stocks fell 0.46 percent Wedne
traders await a deal in
will reopen the gov-
devastating debt
Index

Appendix-1

Assumptions for Comparative Chart in Slide 3

Adjusted Assumptions (Year-3 per merchant)

- **RuPayPay (VC path):** Outstanding loan ~₹40k, Loanbook/GMV = 6.4% (given).
- **BharatPe:** Stronger credit uptake — Outstanding loan ~₹55k, Loanbook/GMV ≈ 7%.
- **Paytm:** Broader base, weaker SME credit uptake — Outstanding loan ~₹25k, Loanbook/GMV ≈ 5%.

Insights of the chart-

- RuPayPay VC path: mid-level intensity, GMV ₹25k cr and loanbook ₹1,600 cr.
- BharatPe: stronger SME focus → higher loanbook per merchant, so ₹2,200 cr loanbook at same base.
- Paytm: lower SME lending penetration → smaller loanbook (₹1,000 cr) despite big consumer payments volume.
- This shows why specialization matters: RuPayPay can realistically aim for BharatPe-like intensity (₹50k+/merchant) in future, but it already outperforms Paytm's SME depth.

Assumptions for Comparative Chart in Slide 3

10 Cities by Tier 2/Tier 3 Merchant & Wholesaler Concentration

1. Jaipur
2. Lucknow
3. Ghaziabad
4. Surat
5. Coimbatore
6. Ludhiana
7. Indore
8. Kanpur
9. Kochi
10. Vijayawada

Ranking by Rebound in Spending, Borrowing, and Investment

- Cities such as Jaipur, Lucknow, Surat, Ludhiana, Indore, and Ghaziabad show the fastest increases in credit/borrowing activity, merchant spending, and investment as the economic cycle turns upward.

Sources-

- https://blog.privatecircle.co/tier-2-momentum-tier-1-muscle-how-indias-startup-geography-is-shifting/
- https://gokiwi.in/blog/indian-cities-with-highest-credit-card-spend-in-2025/

Appendix-2

Table 1- Current Contraction in Economy (Reasons)

Indicator	Direction	Deduction
Interest Rates	Rising	Central banks increase rates to combat inflation, which improves profitability for banks in the financial sector.
Inflation Rate	Increasing / Stable at	A higher rate of inflation occurs at the peak, as demand catches up to and exceeds
GDP Growth Rate	High, but slowing	GDP growth remains positive but is moderating from the high-growth expansionary phase, signaling that the
Consumer Spending &	Slowing	As inflation and interest rates rise, consumers pull back on spending, especially
Business Investment	Strong but potentially at a peak	Corporate balance sheets are healthy, with manufacturing showing readiness for new capital expenditure. However, as the peak is reached, businesses may begin to reassess future spending.
Employment	Steady/Late-cycle increase	Employment growth often stays strong through the peak but may lag slightly, with

Table 2- Cost Breakdown (Integrating low cost Incurred in Model Inspired from China)

Infrastructure Component	Description	Estimated One-Time Cost	Estimated Annual Cost
Automated Underwriting Engine	Develop or purchase a custom AI/ML-driven underwriting engine. ntinuous refinement.	₹5–15 crore (Build) or ₹2–5 crore (SaaS/Platform)	₹1.5–5 crore (Platform fees, API usage, model
Data Infrastructure	Scalable and secure database for storing and processing large	₹2–5 crore	₹1–3 crore (Cloud hosting, data
Payment Gateway Integration	Upgrading your existing payment gateway to access	₹1–3 crore	Included in transaction fees
Compliance and Security	Ensure compliance with RBI regulations, NPCI standards,	₹1–2 crore	₹0.5–1 crore
Marketplace Development	If you build an in-house marketplace, this is the cost for	₹5–10 crore	₹2–4 crore (Maintenance,
Team Expansion (Tech)	Hire specialized data scientists, machine learning engineers, and security experts	₹2–4 crore (Hiring)	₹3–6 crore (Salaries, training)
Total (Low End)		₹16 crore	₹8 crore
Total (High End)		₹39 crore	₹19 crore

WEEKLY CLOSES
TOPIX
-0.69
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Thank You!